Heterogeneity Under Competition

Steven A. Lippman, Kevin F. McCardle[1] and Richard P. Rumelt[1]

[1]Anderson School of Management, UCLA

Abstract

A standard prediction of neoclassical microeconomics is that with perfect competition, free entry, and atomistic firms producing a homogeneous product, equilibrium finds all firms employing that technology which has minimum average cost. That competition drives out inefficient producers and reduces heterogeneity within industries is, consequently, a commonplace rule of thumb in economic thinking. This paper demonstrates that demand uncertainty is sufficient to produce heterogeneity in the equilibrium employment of production technologies and to permit the coexistence of producers exhibiting different minimum average costs. This heterogeneity is ubiquitous because the conditions for its presence are not stringent.

This working paper was subsequently published as:

Lippman, Steven A., Kevin F. McCardle, and Richard P. Rumelt. "Heterogeneity Under Competition." *Economic Inquiry* 29, no. 4 (1991): 774–782.

I. Introduction

One of the features of the world of business, a feature which sets it apart from the world of textbook microeconomics, is sustained heterogeneity among competitors. Whereas neoclassical microeconomic theory maintains that efficient firms (those with lower minimum average cost) will drive out the less efficient firms, casual empiricism demonstrates otherwise. Modern economies are replete with industries in which producers of apparently similar products employ different production technologies or, more generally, different business strategies. Moreover, the differences among technologies or approaches to business persist over long periods of time. For example, some diesel engine manufacturers utilize computer-controlled factories to produce their goods while others employ third-world labor forces. Telecommunications firms may variously employ satellite, microwave, and underground fiber optics channels, and where one airline may commit an older DC-9 to a route, another airline may assign a newer Boeing 757 to the same route.

Heterogeneity in production methods may, of course, stem from technological progress in durable capital goods. [1] In addition, there is a school of thought which ascribes heterogeneity among

competitors to bounded rationality or limitations on imitability. For example, Nelson and Winter [1982] stress the importance of organizational routines in limiting imitation and the diffusion of best practice. Demsetz [1973] argues that competitors may lack the knowledge to copy the techniques of industry leaders, and Lippman and Rumelt [1982] show that even in the absence of information asymmetries among competitors, ambiguity as to how efficiencies are obtained can make it rational for a firm to halt imitative attempts far short of the attainment of best existing industry practice.

II. Background

There is a large body of literature concerning the firm's choice of technology and output under conditions of uncertainty. However, to our knowledge nowhere else has it been demonstrated that in equilibrium uncertainty can lead to heterogeneity of firm types. The effect of risk aversion and demand uncertainty on the behavior of the competitive firm has been analyzed by a number of authors starting with Baron [1971], Leland [1972], and Sandmo [1971]. (Relatedly, Appelbaum and Lim [1982b] demonstrate that the reduction in output induced by uncertain demand and risk aversion is smaller for a monopolist than for a competitive industry.) In extending this literature by allowing free entry, Appelbaum and Katz [1986] demonstrate that the dependence of com-parative-static results upon assumptions concerning absolute and relative risk aversion atrophies and all but disappears. Holthausen [1976] (with a critique by Mayer [1978]) considers a two-stage process whereby the firm installs fixed capital prior to demand being realized and determines optimal variable inputs thereafter. While this two-stage aspect of his model is similar in spirit to our model, there too the focus is upon the role played by the firm's attitude toward risk.

In the equilibrium models of Sheshinski and Drèze [1976], Lippman and McCall [1981], and Appelbaum and Lim [1982a] there is only one possible technology; our model allows for a continuum of technologies. As per Sheshinski and Drèze and Lippman and McCall, but in contrast to Appelbaum and Lim, firms need not commit themselves to production decisions prior to the resolution of uncertainty. Naturally, however, capital must be in place before demand is announced; otherwise firms face no uncertainty.

Appelbaum and Lim [1985] present a model in which a monopolist produces prior to the resolution of uncertainty at a constant marginal cost which is lower than the constant marginal cost of the competitive production which occurs after demand is announced. The ex ante production acts as an entry-deterring precommitment of the monopolist, while demand uncertainty imparts value to the less efficient, more flexible production which occurs after demand is realized. Thus, their paper studies the tradeoffs between commitment and flexibility.

III. Model

We assume that a production technology or firm type is specified by a pair (F, v) where F is the fixed cost and v the constant variable cost of production. Each firm has a capacity constraint on production, assumed without loss of generality equal to one unit. Thus, the cost $c(q)$ of producing q units of the good is given by

$$c(q) = \begin{cases} F + vq & \text{if } 0 \leq q \leq 1; \\ \infty, & \text{if } q > 1. \end{cases}$$

The firm's average unit cost declines in q and the minimum average cost, $C = F + v$, occurs at the capacity constraint.

Industry demand, which is unknown ex ante, is given by the product

$$X \cdot D(p),$$

where X is a random variable whose distribution is known and $D(p)$ is a downward sloping function of price, with $D(p) > 0$ for all p. As opposed to most other models of production in the face of uncertain demand, we do not require production to occur before the uncertainty is resolved (i.e., the value of X is announced). Rather, we require that capacity be installed before demand is announced. In a single period model without inventory, this is tantamount to requiring that the fixed costs be paid before demand is realized. This sets an upper limit on the amount of production, but no preproduction is allowed as a precommitment.

Firms behave as price takers in a competitive environment. Hence, an individual firm will produce only if the price p exceeds the variable cost v of production. This implicitly defines a firm's upward sloping supply curve $s(p)$:

$$s(p) = \left\{ \begin{array}{l} 0 \text{ if } p < v 1 \text{ if } p \geq v \end{array} \right.$$

Firm profits π are given by

$$\pi = pq - c(q) = ps(p) - c[s(p)] \quad = ps(p) - F - vs(p)$$

Substituting in from (2) yields

$$\pi(p) = \begin{cases} -F & \text{if } p < v \\ -F - v + p & \text{if } p \geq v \end{cases}$$

Because these firms are price takers, their supply curves aggregate by addition: n firms of type (F, v) together provide $ns(p)$ units of output. Furthermore, this collective of n firms behaves just like a single price-taker of type (nF, v) with a capacity constraint n. Thus, we are able to implement the

assumption of atomism simply by removing the restriction that the number n of firms be integral.

Consumers are passive and play no strategic role. Their welfare is determined solely through the demand function. The actual demand is unknown to the firms prior to their entering the industry via the installation of capacity, though the distribution of demand is known. Thus, the sequence of events begins with firms entering the industry by installing capacity and paying the fixed cost F; entry continues until each firm type has zero expected profit. Next, demand is announced. Finally, depending on the (auctioneer's) resulting price, firms make their production decisions. Of course, each firm either produces at capacity or produces nothing.

IV. The Ubiquity of Heterogeneity

Suppose there are two types of firms: type A with costs (F_A, v_A), and type B with (F_B, v_B). Without exception, we assume that type A has higher fixed costs but is the more efficient producer: $F_A > F_B$ and $C_A < C_B$ where $C_A = F_A + v_A$ and $C_B = F_B + v_B$. When these inequalities hold, we say that A and B are reciprocal technologies. Clearly, $v_A < v_B$ for reciprocal technologies.

If demand were known, then only type A firms would enter, and entry would continue until price equalled C_A. However, if demand is uncertain, there may be cases where both types of firms coexist in equilibrium. For simplicity, assume that the intensity X of demand can take on only two values: x_l with probability θ_l and x_h with probability $\theta_h = 1 - \theta_l$. High demand intensity x_h leads to price p_h while low demand intensity leads to price p_l with $p_h \geq p_l$. This may be slightly counter to a first intuition. Observed demand is not shifting along one curve, in which case demand rises as price falls; rather, the shift is between two different demand curves, and moving from the low demand curve to the high demand curve opens opportunities for the less efficient firms to produce and market their goods, ultimately driving prices up.

Under what conditions will both types of capacity be installed? It is useful first to establish that any such equilibrium must have the two types following different production strategies. To see this, note that capacity will be installed only if the firm produces in at least one of the two demand states. Because $p_h \geq p_l$, a firm that produces in the low demand state also produces in the high demand state. If a type A firm produces in both demand states, then its expected profit is

$$\mathrm{E}\left(\pi_A\right) = \bar{p} - C_A,$$

where $\bar{p} = \theta_l p_l + \theta_h p_h$ is the expected price. Assuming risk neutrality, it is clear that equilibrium requires $\bar{p} = C_A$. But because $C_A < C_B, E\left(\pi_B\right)$ would be negative if type B firms produce in both demand states. Similarly, production by both types only in the high demand state is inconsistent with zero expected profit. As long as production plans are symmetric with respect to uncertainty, only the most efficient firm type can be present. Because type A firms will always find it profitable to produce if type B does, it must be the type B firms which cease production in the low demand

state. Thus, we have the following result.

OBSERVATION 1. An equilibrium in which both types of capacity are installed must have type A firms producing in both states and type B firms producing only in the high demand state.

We now can be more precise about the conditions enabling an equilibrium containing both types. Type A firms produce in both states of the world. Hence,

$$p_l \geq v_{A'}$$

else type A firms would cease production in the low demand state. As type B firms do not produce in the low demand state, it must be that

$$p_l < v_B.$$

The expected profit of both type A and type B firms is zero so that

$$\bar{p} = C_A$$

and

$$\theta_h p_h = F_B + \theta_h v_B.$$

Finally, the numbers n_A and n_B of type A and type B firms present in equilibrium must satisfy the market clearing conditions:

$$x_l \cdot D(p_l) = n_A$$

and

$$x_h \cdot D(p_h) = n_A + n_B.$$

How difficult is it to satisfy all of these conditions? Our plan is to determine how much freedom is available in the specification of the technologies, demand function, mix of firms, and uncertainty structure X. To begin, take any pair of reciprocal technologies (F_A, v_A) and (F_B, v_B) with total capacities n_A and n_B, respectively. Solving the zero profit conditions (6) and (7) for p_1 yields

$$p_l = v_B - (C_B - C_A)/\theta_l.$$

Because $C_B > C_A$, it is clear upon inspection of (9) that any p_i satisfying the zero expected profit conditions also satisfies (5), ensuring that type B firms shut down when demand is low. In addition, note that whereas (9) ensures $p_l < v_B$, condition (7) implies $p_h > v_B$. Thus, $p_h > p_l$, as is necessary, and type B firms produce in the high demand state.

Using (9), condition (4) becomes $v_B - (C_B - C_A)/\theta_l \geq v_A$, which can be written as

$$\theta_l \geq 1 - (F_A - F_B) / (v_B - v_A).$$

Because the right-hand side is strictly less than unity and θ_l is a probability (whence it is between 0 and 1), there is clearly a continuum of θ_l which satisfy the equilibrium conditions. Thus, as long as θ_l satisfies (10), the prices satisfying the zero expected profit conditions will also guarantee that type A firms produce in both states.

Given a value of $\theta_l < 1$ satisfying (10), there is a unique pair of prices p_l and p_h specified via (6) and (7). Given these prices, x_l and x_h are determined from (8): $x_l = n_A/D(p_l) < (n_A + n_B)/D(p_h) = x_h$ where the inequality follows from $p_h > p_l$ and $D(p)$ downward sloping. The foregoing argument has proven the following:

PROPOSITION 1. Given a pair of reciprocal technologies, a downward sloping demand function $D(p)$, and two positive numbers n_A and n_B, there exists a random variable X with two-point support such that the requisite numbers n_A and n_B of both types of firms exist in the unique equilibrium.

The power of proposition 1 is twofold. Not only does it assert that heterogeneity in equilibrium is possible, it also asserts that no matter how stringently the reciprocal technologies, demand curve, and amounts of type A and type B capacity are specified, we can always find an uncertainty structure X such that both types are present in equilibrium. The next proposition proves the converse: given a random demand, there is an equilibrium which supports both types of firms.

PROPOSITION 2. Given a random variable X with two-point support and a downward sloping demand function $D(p)$, there exists a pair of reciprocal technologies such that both types exist in equilibrium. [2] 2. Actually, there is a continuum of pairs of reciprocal technologies that can exist in equilibrium with a given random demand structure. The proof of proposition 2 solves for one endpoint of the continuum.

Proof. Pick any p_l, p_h pair, with $p_h > p_l$, such that $x_h D(p_h) > x_l D(p_l)$. Set $v_A = p_l F_A = \theta_h (p_h - p_l)$, $F_B = \theta_h (p_h - p_l)/2$, and $v_B = (p_h + p_l)/2$. This specification satisfies (4)-(8): both types install capacity, have zero expected profits, and the market clears.

Together propositions 1 and 2 establish the ubiquity of heterogeneity with uncertain demand. The simplest of conditions is enough to generate the heterogeneity, and the equilibrium distribution of types is not unique. The most basic uncertainty supports a diversity of heterogeneous equilibria.

We also contend that the result of proposition 1 is robust to the assumption of risk neutrality. Adding risk aversion to our model strengthens the result of proposition 1 and supports our contention. Specifically, let $u_i(\pi)$ be the utility associated with profit level π for the entirety of type i capacity, $i = A$ or $i = B$; type i firms act as a single firm in regard to risk. Our method of proof requires the utility functions satisfy the following regularity condition: u' is continuous at 0 and $u'(0) > 0$.

COROLLARY 1. Suppose A and B are reciprocal technologies and $u_A = u_B = u$. If both technology types exist in equilibrium when u is linear, then both types exist in equilibrium when u is concave.

Proof: Let p_h^u and p_l^u denote the equilibrium prices when u is concave and not linear. If the equilibrium were to entail no type A firms, then $p_h^u \geq p_l^u \geq v_B$, so type B firms produce in both demand states. Because type A firms are more efficient, the profit of a type A firm would exceed that of a type B firm in both demand states, whence the expected utility for a type A firm would exceed that of a type B firm. Hence, type A firms will enter.

Suppose there are no type B firms in equilibrium. We claim $p_h^u > p_h$. By assumption $n_B^u = 0$. If $n_A^u \geq n_A + n_B$, then $p_l^u < p_l$ and $p_h^u \leq p_h$, so that type A firms would have negative expected profits. Thus, $n_A^u < n_A + n_B$ and, in turn, $p_h^u > p_h$ as claimed. Consequently, there would be a higher price and hence a higher profit to type B firms in the high demand state; the profit is at least $-F_B$ in the low demand state. As $u'(0) > 0$ and u' is continuous at zero, it follows (see theorem 1 of Lippman and Mamer [1988]) that the expected utility for a sufficiently small value of n_B is strictly positive. □

One might wonder whether it is the discrete nature of the available technologies which produces the existence result of proposition 1. To test this idea, define technology α by (F_α, v_α), where $F_\alpha = \alpha F_A + (1-\alpha)F_B$ and $v_\alpha = \alpha v_A + (1-\alpha)v_B$, and suppose that in addition to the reciprocal technologies A and B, technology α is available for each α in $(0,1)$.

COROLLARY 2. Given an equilibrium with $v_A < p_l$ as per proposition 1, the availability of these alternative technologies leaves the equilibrium unperturbed, for $E(\pi_\alpha) < 0$ for all α.

Proof: First observe that technology α produces only at the high price, for otherwise

$$\mathrm{E}(\pi_\alpha) = -(F_\alpha + v_\alpha) + \theta_h p_h + \theta_l p_l \quad < -(F_A + v_A) + \theta_h p_h + \theta_l p_l = \mathrm{E}(\pi_A) = 0$$

$$\begin{aligned}
\mathrm{E}(\pi_\alpha) &= -F_\alpha + \theta_h (p_h - v_\alpha) \\
&= -[\alpha F_A + (1-\alpha)F_B] \\
&\quad + \theta_h [\alpha (p_h - v_A) + (1-\alpha)(p_h - v_B)] \\
&= \alpha [-F_A + \theta_h (p_h - v_A)] \\
&\quad + (1-\alpha)[\mathrm{E}(\pi_B)] \\
&= \alpha [\mathrm{E}(\pi_A) + \theta_l (v_A - p_l)] < 0.
\end{aligned}$$

(If $p_l = v_A$, then $\mathrm{E}(\pi_\alpha) = 0$ and type α is viable, but $p_l < v_A$ provided θ_l is selected to ensure the inequality in (10) is strict.) □

V. Comparative Statics

Whereas only the most efficient firms can survive when there is no uncertainty, proposition 1, along with corollaries 1 and 2 , reveals that demand uncertainty is sufficient to produce heterogeneity in the employment of production technologies: producers exhibiting differing minimum average costs

coexist in equilibrium. The presence of the less efficient technology in equilibrium leads us to inquire in this section whether uncertainty per se has a deleterious impact upon social welfare.

Because the industry is competitive, consumer and social welfare are equivalent. As there are two possible demand states, the expected consumer surplus [3] S is given by (11) $S = \theta_l \int_{p_l}^{\infty} x_l D(p)dp + \theta_h \int_{p_h}^{\infty} x_h D(p)dp$.

The next proposition demonstrates that uncertainty per se is deleterious: an increase in uncertainty reduces the expected

Next, observe that $v_A - p_I < 0$ whence 3. This measure is a well-behaved social ordering as long as the uncertainty concerns some exogenous event and not the intensity of buyer preferences. consumer surplus. We alter the riskiness of X via dilation:

$$Y_\delta = \delta[X - \mathrm{E}(X)] + \mathrm{E}(X)$$

For values of $\delta > 1$, this alteration of X results in the familiar mean-preserving increase in risk; for $\delta < 1$ the result is a meanpreserving decrease in risk.

PROPOSITION 3. Let a pair of reciprocal technologies, a pair $(p_l p_h)$ of prices with $p_l < p_h$ a downward sloping demand function $D(p)$, and a random variable X with two-point support be given such that equilibrium supports both technologies at the prices $(p_b p_h)$, and replace X by the dilation Y_δ. Then an increase in δ leads to (i) a decrease in n_A and an increase in $n_A + n_B$, and (ii) a decrease in S.

Proof: From the market clearing conditions given by (8) (substituting in Y_δ for X), an increase in δ leads to a decrease in n_A and an increase in $n_A + n_B$. Hence, the proportion of the market taken up by the less efficient firm increases with δ. Observing that $dY_\delta/d\delta = X - E(X)$ and also that p_l and p_h do not vary with δ, we easily differentiate S as follows:

$$\begin{aligned} dS/d\delta &= \theta_l \int_{p_l}^{\infty} [x_l - \mathrm{E}(X)]\, D(p)dp \\ &\quad + \theta_h \int_{p_h}^{\infty} [x_h - \mathrm{E(X)}]\, D(p)dp \\ &< [\theta_l\, [x_l - \mathrm{E}(X)] + \theta_h\, [x_h - \mathrm{E}(X)]\} \\ &\qquad \int_{p_h}^{\infty} D(p)dp = 0 \end{aligned}$$

An increase in uncertainty reduces social welfare and increases both the number and proportion of the less efficient type B firms. However, the next proposition confirms that consumers would be even worse off were the type B technology not available.

PROPOSITION 4. Given a pair of reciprocal technologies, a downward sloping demand function $D(p)$, and a random variable X with two-point support such that both types exist in equilibrium, then the expected consumer surplus is smaller under the regime which excludes type B firms.

Proof: Denote the high and low prices and the number of type A firms in the market under the regime which excludes type B firms by p'_h, p'_l, and n'_A, respectively. Under the regime which allows type B firms, denote the equilibrium prices and number of firms by p_h, p_v, n_A, and n_B. It is important to observe that $n'_A > n_A$ and, especially, $n'_A < n_A + n_B$. (If $n'_A \geq n_A + n_B$, then $p'_h \leq p_h$ and $p'_l < p_l$ whence the expected profit for type A is negative.) Consequently, $p_h \langle p'_h$ and $p_l \rangle p'_l$. Thus,

$$\begin{aligned} S(p_l, p_h) - S(p'_l, p'_h) &= \theta_h \int_{p_h}^{p'_h} x_h D(p)\, dp - \theta_h \int_{p'_l}^{p_l} x_l D(p)\, dp \\ &> \theta_h n'_A (p'_h - p_h) - \theta_l n'_A (p_l - p'_l) \\ &= n'_A \{(\theta_h p'_h + \theta_l p'_l) - (\theta_h p_h + \theta_l p_l)\} = 0, \end{aligned}$$

where the inequality holds because $x_h D(p) > n'_A$ for $p < p'_h$ and $x_l D(p) < n'_A$ for $p > p'_\nu$ and the last equality follows from (6).

VI. Discussion and Conclusions

In a standard competitive setting with price-taking atomistic firms, demand uncertainty is sufficient to induce heterogeneity in the production technologies or firm strategies that are present in equilibrium. Although the setting analyzed here is stylized and very simple, the results are not restricted to this setting. In particular, the linearity of the production function specified in (1) is not required to ensure a heterogeneous equilibrium; cost functions more complex than these also can yield heterogeneous equilibria. In addition, it is possible to extend our analysis to the general case of k technologies by letting X have k-point support.

The results obtained here imply that the static concept of efficiency may be grossly misleading when applied to a context within which demand is uncertain. For example, because a cartel often enfranchises a fringe of inefficient producers (e.g., the Swiss watch industry in the 1950s), industrial organization economists sometimes take the presence of many inefficient producers as evidence that efficient producers have restricted output or capacity additions. Proposition 4 reveals that "inefficient" fringe producers can be the efficient solution to demand uncertainty.

Another example concerns the historic role of American Motors in the U.S. automobile industry. For many years American Motors maintained a position of relatively low fixed costs but relatively high variable costs. This position corresponds to the less efficient type B production technology in our model. The standard static view leads to recommendations for exit, renovation, or merger (American Motors was acquired by Chrysler Corporation in 1987). However, assuming that its technology choice was not driven by special product characteristics, our model suggests that a company like American Motors might be part of the efficient solution to the problem of providing "peak-load" capacity in a world with uncertain demand. This view leads to an alternate set of

recommendations such as further reducing fixed costs and crafting special labor contracts promoting flexibility.

References

Appelbaum, E. and E. Katz. "Measures of Risk Aversion and Comparative Statics of Industry Equilibrium." American Economic Review 76, 1986, 524-29.

Appelbaum, E. and C. Lim. "Long-Run Industry Equilibrium with Uncertainty." Economic Letters 9, 1982a, 139-45.

____ "Monopoly Versus Competition Under Uncertainty." Canadian Journal of Economics 15, 1982b, 355-63. Baron, D. "Demand Uncertainty in Imperfect Competition." International Economic Review 12, 1971, 196-208.

Demsetz, H. "Industry Structure, Market Rivalry, and Public Policy." Journal of Law and Economics 16, 1973, 1-9.

Holthausen, D. "Input Choices and Uncertain Demand." American Economic Review 66, 1976, 94103.

Leland, H. "Theory of the Firm Facing Uncertain Demand." American Economic Review 62, 1972, 27891.

Lippman, S. A. and J. W. Mamer. "When Many Wrongs Make a Right: An Asymptotic Analysis of Risk Aversion and Additive Risks." Probability in Engineering and Information Sciences 2, 1988, 115-27.

Lippman, S. A. and J. J. McCall. "Competitive Production and Increases in Risk." American Economic Review 71, 1981, 207-11.

Lippman, S. A. and R. P. Rumelt. "Uncertain Imitability: An Analysis of Interfirm Differences in Efficiency Under Competition." Bell Journal of Economics 13, 1982, 418-38.

Mayer, W. "Comment." American Economic Review 68, 1978, 231-32.

Nelson, R. R. and S. Winter. An Evolutionary Theory of Economic Change. Cambridge MA.: Belknap, 1982.

Sandmo, A. "On the Theory of the Competitive Firm Under Price Uncertainty." American Economic Review 61, 1971, 65-73.

Sheshinski, E. and J. H. Drèze. "Demand Fluctuations, Capacity Utilization, and Costs." American Economic Review 66, 1976, 731-42.